



University
of Exeter

BEMM828 - MSC PROJECT PROPOSAL

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1. TOPIC

Dynamic Markdown Optimisation: Predictive Analytics for Perishable Inventory Management in Grocery Retail

2. INTRODUCTION

Food wastage is currently considered one of the main problems of an ethical and operational nature encountered by the food retail business. According to the statistics provided by the FAO, more than one-third of the global supply of edible products goes to waste. An overwhelmingly large share of that number falls upon the stage of retail when products become outdated. In the UK alone, food retailers create about 270,000 tons of food waste per annum. Such waste primarily comes from perishable goods that do not sell in time such as dairy, bakery, and produce (Wrap, 2022).

One of the most efficient ways to deal with this problem is the use of a markdown strategy, meaning the reduction of price to speed up product sales. It is estimated that in the case of proactive measures, food retail companies could make significant financial gains and reduce the amount of waste to zero. These guesses completely disregard the immense amount of information available to us through advanced checkout and inventory systems. Due to this, reductions in prices occur either too late or not reduced by enough, resulting in failure to sell them and more waste. Recent advancements in machine learning algorithms and predictive analysis techniques provide a solution to turn such intuition into a well-thought-out procedure. Analyzing correlations between sales and variables such as timing of a markdown, its scale, as well as the time of a day, day of the week, product categories, inventories, and even weather conditions allow developing models calculating the absolute optimal schedule for markdowns (Ferreira et al., 2016).

Consequently, the goal of the proposed study is to understand the impact of time of a day, available inventories, different kinds of products, as well as ongoing promotions on perishables selling rate. This research will focus on creating a machine learning algorithm to generate the most adaptive markdown strategies. Then, the idea is to develop a simple dashboard prototype based on the results obtained with the help of Power BI. In summary, this project provides a great opportunity to contribute not only to academia but also to create a practical tool for retailers.

2.1 Research Questions

Question 1: Are there any factors that really influence sales rates of the goods sold at a discounted price? Is it the time of the day, day of the week, amount of the remaining stock on the shelves, or the nature of the products themselves?

Question 2: Which machine learning models prove to be most precise in their prediction of product sales speed if we change the prices on various discounts?

Question 3: How can these complicated model predictions be turned into understandable information shown in a Power BI dashboard?

2.2. Research Aim and Objectives

AIM- To design and develop a machine learning model that assists grocery chains to discover the optimum times and price points for discounts on fresh products.

OBJECTIVES-

- Conduct the literature review: Identify factors behind the successful sale of food products at reduced prices according to the scientific literature.
- Collect the data: Gather and prepare the actual sales-related statistics.
- Build the models: Test several algorithms of machine learning (e.g., Random Forest and XGBoost) and select the most efficient one for predicting demand.
- Create the visualization dashboard: Construct the prototype of the dashboard using the Power BI software.
- Evaluate the outcomes: Conduct the analysis of efficiency and applicability of the solution within the context of the chosen organization.

3. LITERATURE REVIEW

3.1. Perishable Inventory Management

Perishable inventory management has been one of the most challenging topics in operations management since long ago. The difference between perishable and non-perishable items in the context of sale lies in the limited shelf life of perishable products, the randomness of consumer demand, and the certainty of losses associated with the perishment of such items. The first person who developed a fundamental principle regarding perishable inventory management was Steven Nahmias (1982). He made mathematical proof that ordering and pricing policies for products that perish should be considered independently from the ordering and pricing of non-perishable products. Chen and Sapra (2013) developed the topic of managing perishable items by considering the importance of dynamic pricing in that process. According to them, stocking and pricing of the same product cannot be managed separately, as they should go together to have an efficient store operation policy.

Recently, scientists have turned their attention towards modern supermarkets. According to Kannan and his colleagues (2018), such businesses face numerous problems since they work with multiple products, have huge numbers of clients, and are burdened by exceptionally small profit margins. Indeed, due to such tiny margins, just several wrong steps regarding stock management may make a store fail miserably. What stands out about the findings made thus far is that the demand for products with approaching expiration dates is exceptionally elastic and highly context dependent. People do not resist discounts, yet depending on the type of product, time of day, and reasons to visit the supermarket, they may perceive the price cut in a particular way (Akkaş et al., 2019). Due to the complexity of human behaviour, traditional mathematical models cannot provide accurate estimates. The necessity for demand models capable of dealing with these complex interactions lies in the inherent nature of the problem under discussion, which is why ML solutions can be used successfully.

3.2. Dynamic Pricing and Markdown Optimisation

Dynamic pricing refers to price changes based on current market conditions; this practice is widespread in the context of booking flights or reserving rooms in hotels

(Talluri & Van Ryzin, 2004). The application of this approach in supermarket markdown optimization is a relatively new research direction. Specifically, Avinadav et al. (2014) developed an algorithm for determining optimal markdown plans, showing that price changes depending on the product's residual shelf-life time would minimize losses associated with product expiration and lost revenue. The next step was made by Gabriel Ferreira and his associates (2016) when they carried out a practical experiment of markdown optimization in a large US retail company.

They applied regression analysis and mathematical programming to calculate price recommendations. These authors achieved up to 9% additional revenue with their system compared to existing pricing methods. Nevertheless, they used rather primitive regression models in their experiments. Since then, scientists like Robert Fildes (2022) began to experiment with significantly more sophisticated and non-linear methods of Machine Learning that include Gradient Boosting Trees and Random Forest. The modern approach provides much better forecasting of highly fluctuating demand for goods in retail.

3.3. Machine Learning in Retail Demand Forecasting

The application of machine learning in retail demand forecasting has grown substantially within the last ten years due to increasing transactional data availability and open-source ML platforms. In their literature review on machine learning usage in SCM, Boone et al. (2019) found that ensemble models, especially Random Forest and Gradient Boosting, showed better performance compared to statistical methods in situations where many interacting variables were involved. Markdown clearance prediction is one such case as the dependent variable, in this case, is likely to be impacted by numerous interacting factors. XGBoost, which stands for the efficient implementation of the gradient boosting algorithm, has shown the best performance among all methods used for predicting values using structured tabular data (Chen & Guestrin, 2016). Recently, Punia et al. (2020) have demonstrated the advantages of employing similar techniques for demand forecasting of fresh food products.

3.4. Data Driven Decision Support in Retail Operations

The interpretation of predictive models' results in operational tools is an additional issue which should not be overlooked. In recent years, Power BI and similar business intelligence software have become popular among retailers in their role as the tool layer between analytics and operation decision-making (Delen & Ram, 2018).

Caro and Gallien (2012) described the effectiveness of decision support systems for managing retail inventory through an example of Zara, showing how immediate dashboards could help make better decisions. The general approach to creating efficient operational dashboards, which includes such elements as data visualisation and workflow compatibility, has been defined in the IS field (Few, 2006).

3.5. Food Waste and Sustainability in Retail

In addition to the operational and economic considerations, the issue of perishable markdown optimization is also being positioned in the context of sustainability within the food retail industry. With the UK Government's pledge to reduce food waste by 50% by 2030, which corresponds to the target set out in the UN SDG 12.3, grocery chains face increasing pressure from regulators and public opinion in terms of proving their contribution to achieving these goals (Wrap, 2022; United Nations, 2015).

Research carried out by Papargyropoulou et al. (2014), for example, highlights the role of food waste within the food systems approach, noting that retail-related measures such as pricing can be powerful tools to address waste.

4. RESEARCH DESIGN

4.1. Research Approach

The research adopts a positivist approach that sees customer clearance behaviour and markdown results as objective realities with patterns that can be predicted (Bryman, 2016). The research uses a quantitative approach based on deduction. Existing dynamic pricing and demand forecasting concepts are used to come up with the predictor variables. The approach is perfect for achieving the research objectives since they do not intend to explore the psychology behind the decision-making process of the manager but rather develop predictive models.

4.2. Data Strategy

The study will utilise secondary sources of quantitative data, given that collecting primary quantitative data from grocery retailers is practically impossible in the scope of an MSc project. Using secondary data is not only methodologically sound but also commonly accepted practice in the field of retail analytics (Fildes et al., 2022). The key primary dataset to be used is "The Complete Journey" dataset provided by Dunnhumby (Dunnhumby, 2020). It is openly accessible to scholars for research purposes on Kaggle and not subject to any competition restrictions. Dunnhumby, the company behind the Tesco Clubcard loyalty program, produced the dataset. It contains household-level transactional data from two years of the operations of a grocery retailer. The transactional data include purchase details in relation to perishables (fruit and vegetables, meat and fish, and dairy products), date information about promotional activities, and a separate "Causal" dataset specifying dates when specific products were promoted either individually or included in weekly campaigns. Therefore, the nature of promotions presented in the dataset resembles markdown events and allows studying promotional effect on sales uplift. To augment the Dunnhumby dataset and ensure a more robust testing of the prediction models, the UCI Online Retail II Dataset (Chen et al., 2012) will be used for supplementary research purposes. This openly available dataset, which is part of the UCI Machine Learning Repository, comprises over one million transaction records of an online UK retailer and includes timestamps accurate to the minute (InvoiceDate). The ability to analyse transactions down to the minute level would thus allow time-of-day patterns related to Purchase Behaviour, a core variable in markdown clearance optimization, to be analyzed. As needed, synthetic variables for Markdown Depth and Clearance Rate will be generated through simulation methods from perishable pricing theory (Avinadav et al., 2014).

4.3. Analytical Framework

The entire analytics process involves passing through the following four stages:

- Data Preprocessing: Preparation of raw data and construction of custom features such as inventory level, time interval, day-of-the-week indicator, size of discounts offered, and item age.
- Exploratory Data Analysis: Visualization of distributions and relationships to uncover any latent patterns over time.

- Machine Learning: Implementation and comparison of three different algorithms -Logistic Regression, Random Forest, and XGBoost that will be assessed using cross-validated AUC-ROC and F1 scores.
- Power BI Dashboard: Visual representation of complex analyses in the form of an interactive prototype based on Stephen Few's clarity criteria (2006).

4.4. Analytical Tools

Data processing and modelling will be performed using Python (scikit-learn, XGboost, pandas). Prototyping of dashboards will be performed using Microsoft Power BI Desktop. All code will be well-documented and replicated through a structured GitHub repository.

5. ETHICS

Since the research uses only secondary data sets that are publicly available, there is no use of any personal data, human subjects, or interactions with the organizations under consideration. As such, concerns regarding informed consent, anonymity, and gaining access do not apply in this case, thereby minimizing any ethical risks associated with the study. Nevertheless, it should be recognized that the social consequences of such a research study need careful evaluation. If employed in practice, the algorithms would have consequences for vulnerable consumers purchasing food at discount prices due to their proximity to expiration dates. These consequences would be analysed thoroughly in the final report (University of Exeter, 2021).

6. TIMETABLE/ GRANTT CHART

Task / Activity	Jun	Jul	Aug	Sep
Literature Review & Data Identification	•			
Data Collection & Preprocessing		•		
ML Model Development & Training		•	•	
Power BI Dashboard Design			•	
Analysis & Results Interpretation			•	
Draft Report Writing				•
Supervisor Review & Revision				•
Final Submission				•

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